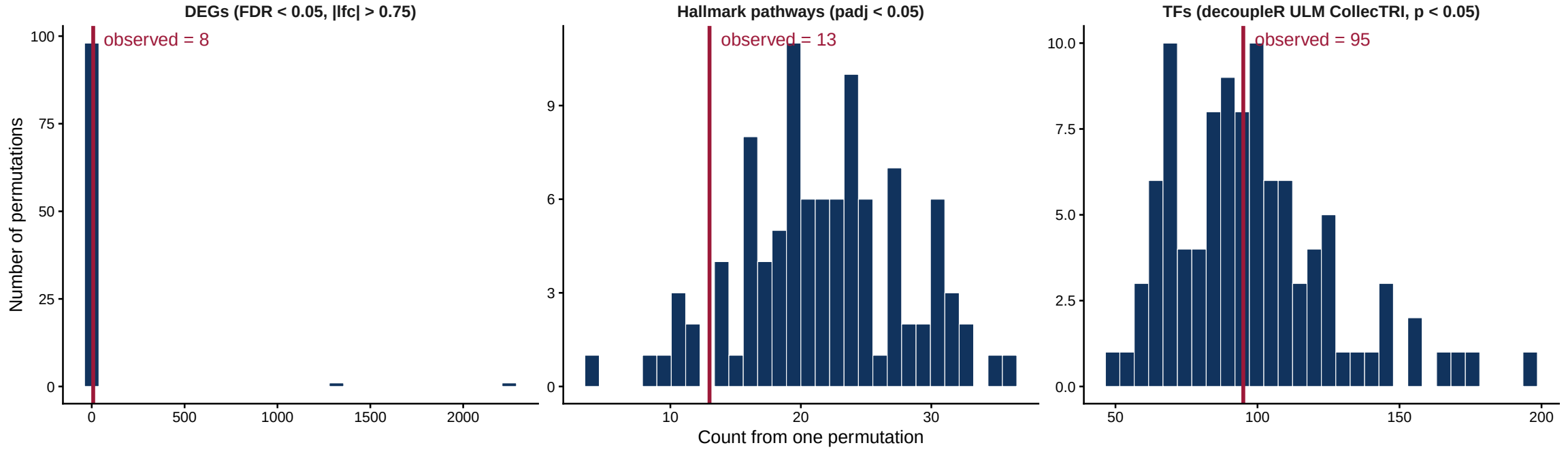
A549 — label−permutation null distribution (N = 100 shuffles)

Treatment labels (LOY / sgCTRL / PARENTAL) shuffled across samples; unpaired DE for both null and observed